Assignment 1

Problem Statement: Build a simple text generator using Markov Chains or any pretrained model

1. Introduction

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on enabling machines to understand and generate human language. One of the simplest approaches to text generation is using **probabilistic models**.
A **Markov Chain** is a mathematical model that predicts the next state based only on the current state, following the **memoryless property**. This property makes Markov Chains suitable for basic text generation tasks where the next word depends only on the current word.

In this assignment, a simple text generator is implemented using a **Markov Chain model**, which learns word-to-word transitions from a given text corpus and generates new text sequences probabilistically.

2. Objective

The main objectives of this assignment are:

- To understand the concept of **Markov Chains** in probabilistic modeling
- To implement a **simple text generator** using word-level Markov Chains
- To learn how text can be generated using **transition probabilities**
- To gain hands-on experience with **basic NLP techniques** without using deep learning

3. Theory

A **Markov Chain** is a stochastic process in which the probability of transitioning to the next state depends only on the current state and not on the sequence of past states.

Markov Property

$$P(X_{n+1} \mid X_n, X_{n-1}, \ldots, X_1) = P(X_{n+1} \mid X_n)$$

In text generation:

- Each **word** is treated as a state
- The **next word** is chosen based on the probability distribution of words that follow the current word in the training text

Working Principle

1. **Training Phase**
    - The input text is split into words
    - A transition model is created where each word maps to a list of possible next words
2. **Text Generation Phase**
    - A starting word is selected

    - The next word is randomly chosen from the learned transitions
    - This process is repeated until the desired text length is achieved

Since the model generates text based on probability, the output may vary each time, making it a **generative model**.

Advantages

- Simple and easy to implement
- Fast and computationally efficient
- Useful for understanding probabilistic text generation

Limitations

- Does not understand grammar or semantics
- Considers only short-term dependencies
- Generated text may be repetitive or incoherent

4. Conclusion

In this assignment, a simple text generator was successfully built using a **Markov Chain model**. The model learned word transition patterns from the training text and generated new sentences based on probabilistic selection.
Although the generated text lacks deep contextual understanding, the implementation effectively demonstrates the core idea of **probabilistic generative modeling** in NLP.
This approach provides a strong foundation for understanding more advanced text generation techniques such as **Recurrent Neural Networks (RNNs)** and **Transformer-based models**.